CIE 3

Total points 16/20

Student's Name and Division *

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✓ 1. Which of the following buses are present in a micro-controller for transferring data from one place to another? 1/1

- ○ data bus only
- ○ data bus, address bus
- ○ address bus only
- ◉ address bus, data bus, control bus ✓

✗ 2. Which of the following micro controller doesn't match with its architecture below? 0/1

- ○ ARM7 – Von Neumann
- ○ Microchip PIC – Harvard
- ◉ ARM9 – Harvard ✗
- ○ MSP430 – Harvard

Correct answer

- ◉ MSP430 – Harvard

✓ 3. When the micro controller executes some arithmetic operations, then the flag bits of which of the following register are *1/1

- ○ DPTR
- ◉ PSW ✓
- ○ PC
- ○ SP

✓ 4. The total space for the data memory available in the AVR-based micro controller is? 1/1

- ○ FFFH
- ○ FH
- ◉ FFFFH ✓
- ○ FFFFFFH

✗ 5. Why are ULN2803 normally used between the micro controllers and the relays? *0/1

- ○ for increasing the voltage capability required by a relay
- ○ for switching purposes
- ○ for increasing the current capability required by a relay
- ◉ all of the mentioned. ✗

Correct answer

- ◉ for increasing the current capability required by a relay

✓ 6. Which of the following bit/s of the status register that allows the micro controller to operate in its low power mode? *1/1

- ◉ CPU off ✓
- ○ Z
- ○ N
- ○ Reserved

✓ 7. Unlike microprocessors, micro controllers make use of batteries because they have *1/1

- ○ high power dissipation
- ◉ low power consumption ✓
- ○ low voltage consumption
- ○ low current consumption

✓ 8. 8051 series has how many 16 bit registers? * 1/1

- ◉ 2 ✓
- ○ 3
- ○ 1
- ○ 0

✓ 9. When 8051 wakes up then 0x00 is loaded to which register? * 1/1

- ○ PSW
- ○ SP
- ◉ PC ✓
- ○ None of the mentioned

✓ 10. When the micro controller executes some arithmetic operations, then the flag bits of which register are affected *1/1

- (●) PSW ✓
- ( ) SP
- ( ) DPTR
- ( ) PC

✗ 11. How many bytes of bit addressable memory is present in 8051 based micro controllers? 0/1

- ( ) 8 bytes
- (●) 32 bytes ✗
- ( ) 16 bytes
- ( ) 128 bytes

Correct answer

- (●) 16 bytes

✓ 12. What is the primary purpose of an interrupt in a computer system? 1/1

- ( ) To stop the execution of the program
- (●) To handle events asynchronously without polling ✓
- ( ) To increase the processing speed
- ( ) To reduce memory usage

✓ 13. What does the Interrupt Descriptor Table (IDT) store? * 1/1

- (●) The locations of interrupt service routines (ISRs) ✓
- ( ) The memory addresses of active tasks
- ( ) The addresses of hardware devices
- ( ) The configuration of system clocks

✓ 14. Which of the following statements is true about enabling and disabling interrupts? *1/1

- ( ) Interrupts are enabled by default and cannot be disabled
- (●) Interrupts can be disabled using the CLI (Clear Interrupt Flag) instruction ✓
- ( ) Disabling interrupts leads to immediate system crashes
- ( ) Enabling interrupts prevents the CPU from performing any other task

✓ 15. What is the purpose of an interrupt priority? 1/1

- ( ) To ensure that all interrupts are handled equally
- (●) To handle more critical interrupts before less important ones ✓
- ( ) To disable all interrupts for system safety
- ( ) To prioritize only hardware interrupts over software interrupts

✓ 16. What is an exception condition in a processor? 1/1

- ○ condition where the CPU automatically handles the exception by restarting
- ◉ An event that forces the system to perform specific error handling or recovery ✓
- ○ A normal event in the system that does not require special handling
- ○ A condition where interrupts are automatically disabled

✗ 17. Which of the following best describes an interrupt service routine (ISR)? 0/1

- ○ A program that handles exceptions by stopping execution
- ◉ A special function that handles interrupts when they occur ✗
- ○ A function that waits for an event to occur
- ○ A system that enables and disables interrupts

Correct answer

- ◉ A function that waits for an event to occur

✓ 18. What is the function of the error code in exception handling? 1/1

- ○ To provide information about the source of the interrupt
- ○ To indicate a successful task execution
- ◉ To represent the exception or error condition encountered ✓
- ○ To disable all further interrupts

✓ 19. While CPU is executing a program, an interrupt exists then it 1/1

- ○ follows the next instruction in the program
- ○ jumps to instruction in other registers
- ◉ breaks the normal sequence of execution of instructions ✓
- ○ stops executing the program

✓ 20. Which of the following interrupt types in 80386 **is** *not maskable***?** 1/1

- ○ INT n
- ◉ NMI ✓
- ○ INTO
- ○ INT3

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